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子計畫一
Hardware-Friendly Highways: From Efficient Dynamic Routing to MoE Architectures

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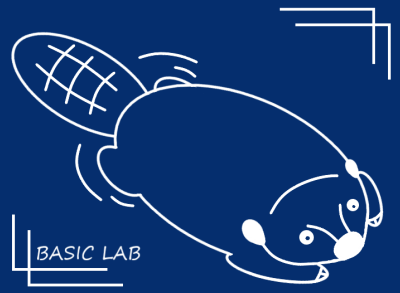
Outline

- Introduction & Motivation
- Sparse Inference for LLMs
  - Dynamic Input Sparsity
  - Dynamic Width Sparsity
  - Dynamic Depth Sparsity
    - Pruning and Early Exit
    - Recurrence and Looped Architectures
    - Dynamic Routing and Modular Inference
- Proposed Research
  - Hardware-Friendly Highway
  - MoE Speculative Decoding

Introduction & Motivation



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Introduction & Motivation

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Bottleneck in LLM inference

- **Computational Intensity**: LLMs require massive FLOPs, leading to high deployment costs.
- **Memory-Bound Nature**: Inference is heavily restricted by memory bandwidth rather than raw compute power.
- **Utilization Gap**: hardware utilization remains low due to data movement overheads.

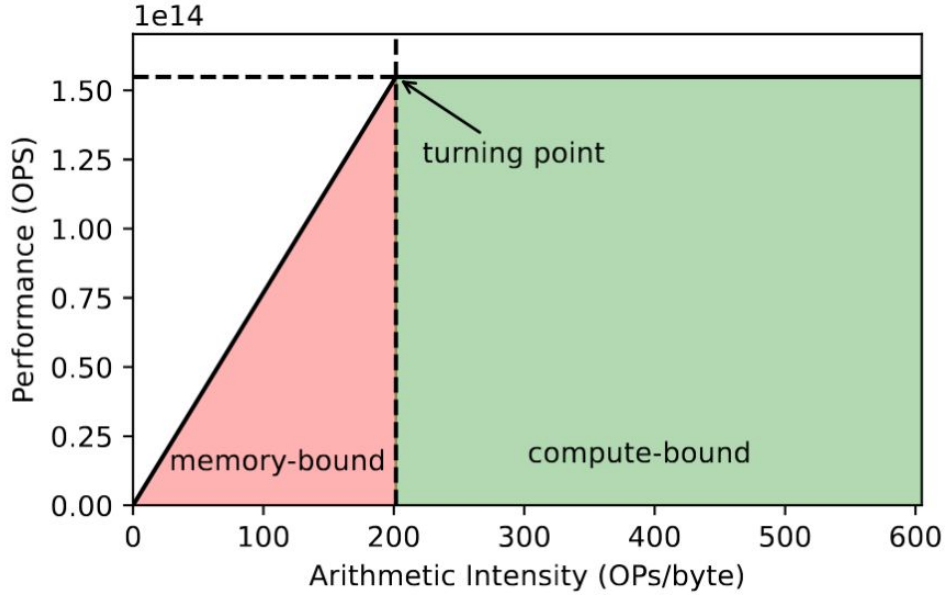

Figure 5. Demonstration of the Roofline model of Nvidia A6000 GPU. The computation is in FP16.

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Introduction & Motivation

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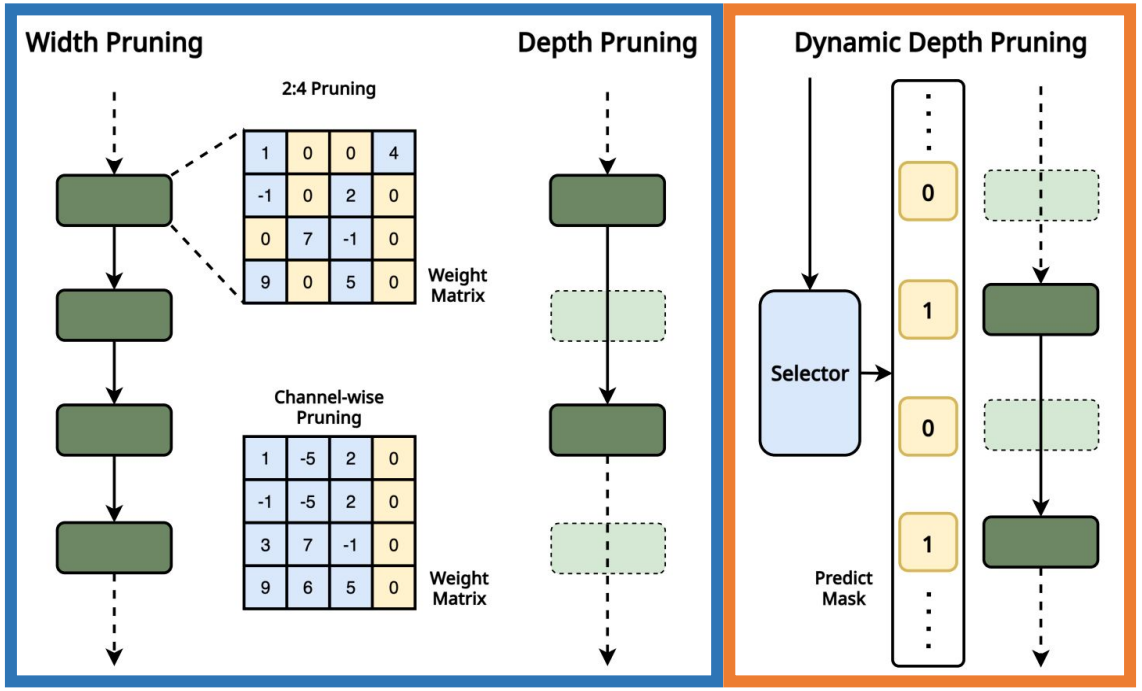
Static Pruning

- Removes redundant parameters permanently.
- **Pain Point**: Often leads to significant **accuracy degradation** and lacks input-dependent flexibility.

Dynamic Routing / Pruning

- Activates specific paths based on input complexity.
- **Pain Point**: Causes **branch prediction failures** and **non-contiguous memory access**, limiting actual speedups on standard hardware.



Common Routing Patterns
&
Hardware-friendly highways

Introduction & Motivation

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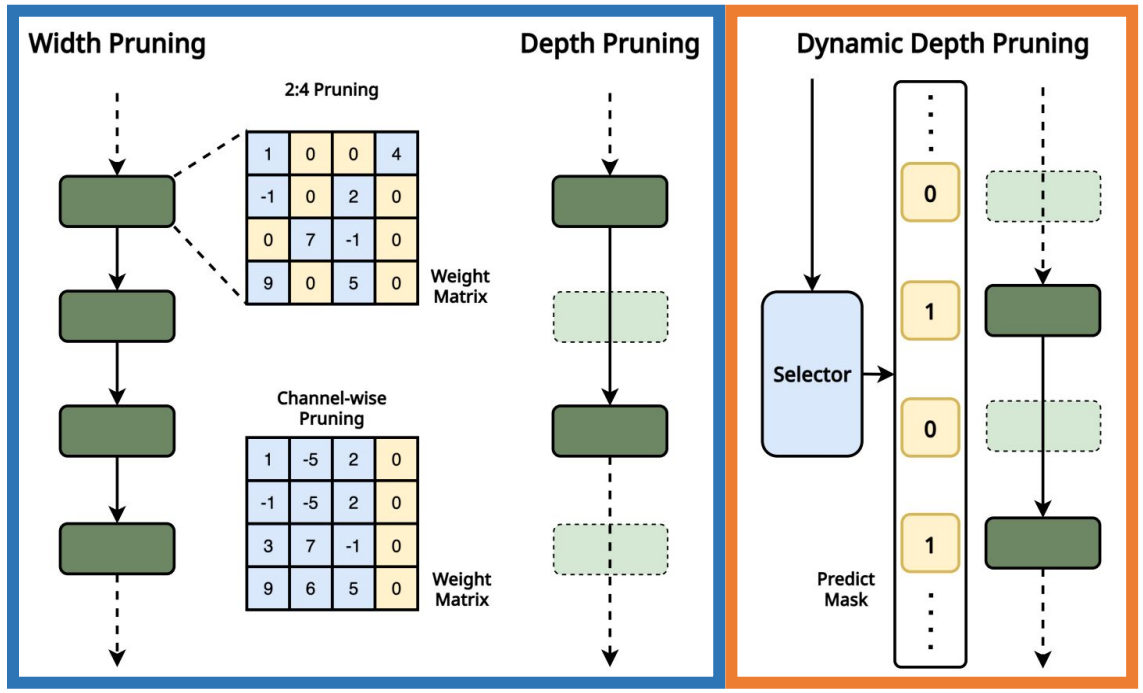
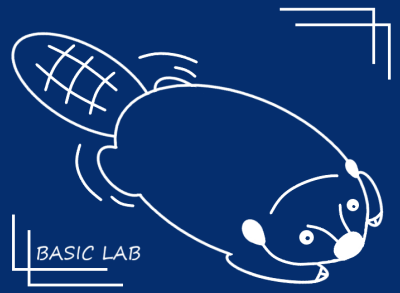


Common Routing Patterns
&
Hardware-friendly highways

- **Efficient Dynamic Routing**: Reducing depth-wise redundancy.
- **Scalable MoE Architectures**: Expanding width-wise capacity without increasing inference cost.

Sparse Inference for LLMs



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Dynamic Input Sparsity

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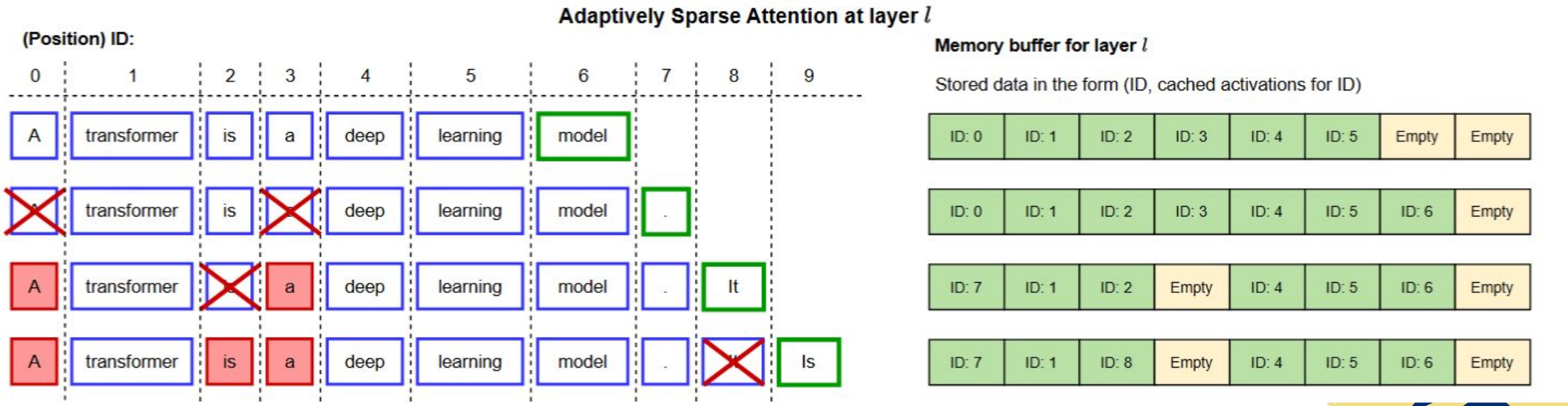
- Token Pruning
- Token Merging
- Dynamic KV Cache Eviction



Dynamic Context Pruning for Efficient and Interpretable Autoregressive Transformers
https://arxiv.org/abs/2305.15805

Dynamic Input Sparsity

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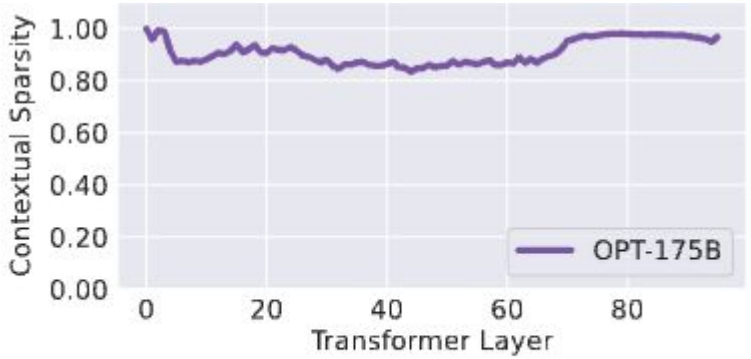
Why input sparsity: contextual sparsity exists and can be predicted.

85% contextual sparsity

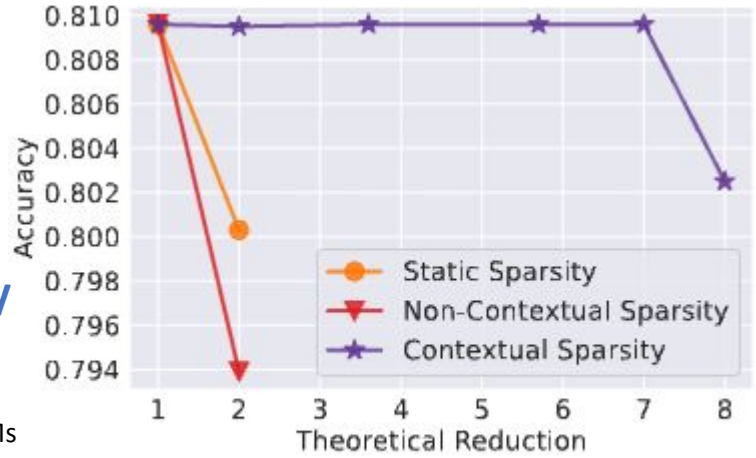


(a) Contextual Sparsity

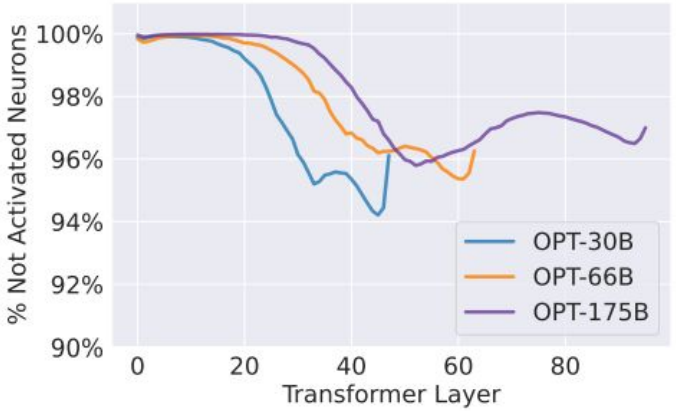
(a) Contextual sparsity in Attention Head

80% sparsity In Attention Head

Parameter Reduction without sacrificing accuracy



(b) Accuracy-Efficiency Trade-offs



(b) Contextual sparsity in MLP Block

95% sparsity in MLP Block

Deja Vu: Contextual Sparsity for Efficient LLMs at Inference Time
https://arxiv.org/abs/2310.17157

Deja Vu: Contextual Sparsity for Efficient LLMs at Inference Time

(PMLR 2023)

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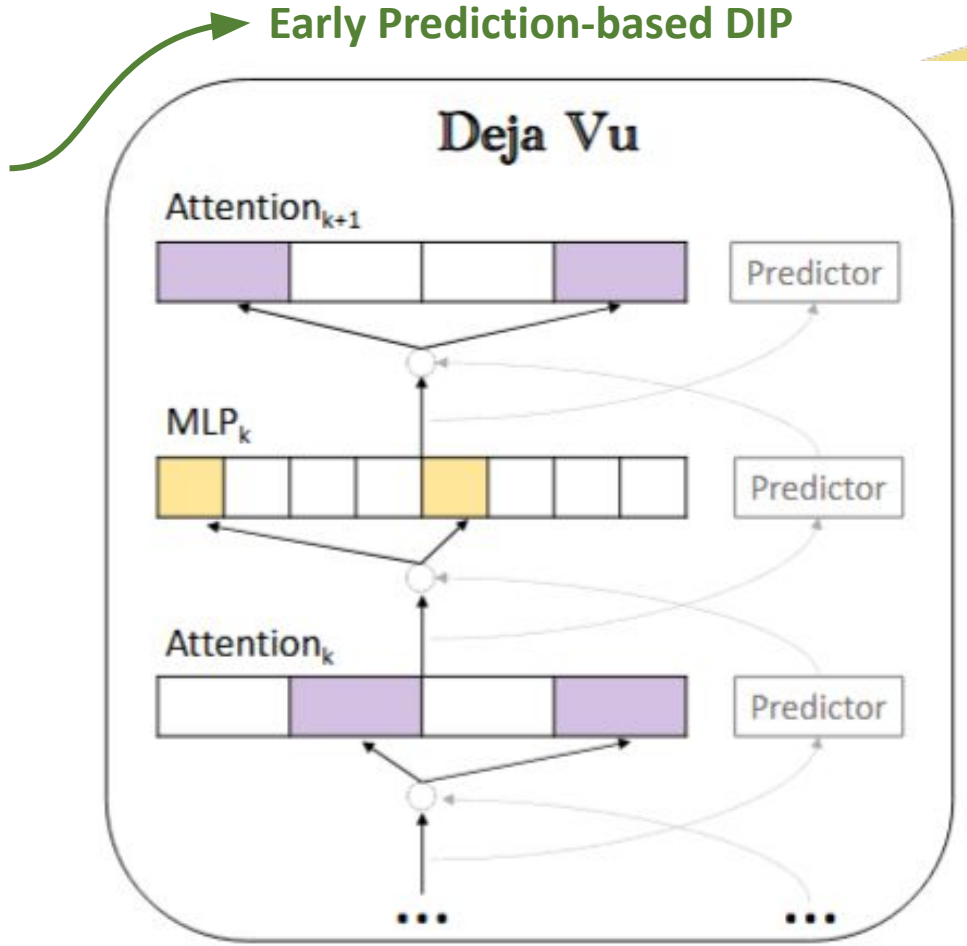
Traditional ReLU-based LLMs

- **High Natural Sparsity**: These models naturally produce a high degree of zero-valued activations.
- **Predictive Pruning**: Small predictors could accurately identify these zeros to enable effective weight pruning.
- **Stable Performance**: This inherent property allowed for acceleration without sacrificing accuracy.

Modern SwiGLU-based LLMs (e.g., Mistral-7B, Phi-3-Medium)

- **Low Natural Sparsity**: SwiGLU networks have very few "hard zeros," making them naturally dense.
- **Unpredictable Patterns**: While they can be pruned based on magnitude, the resulting sparsity patterns are difficult to forecast.
- **The Failure of Predictors**: Traditional predictor-based approaches become ineffective and cause significant accuracy degradation.



Efficient LLM Inference using Dynamic Input Pruning and Cache-Aware Masking

(MLSys 2025)

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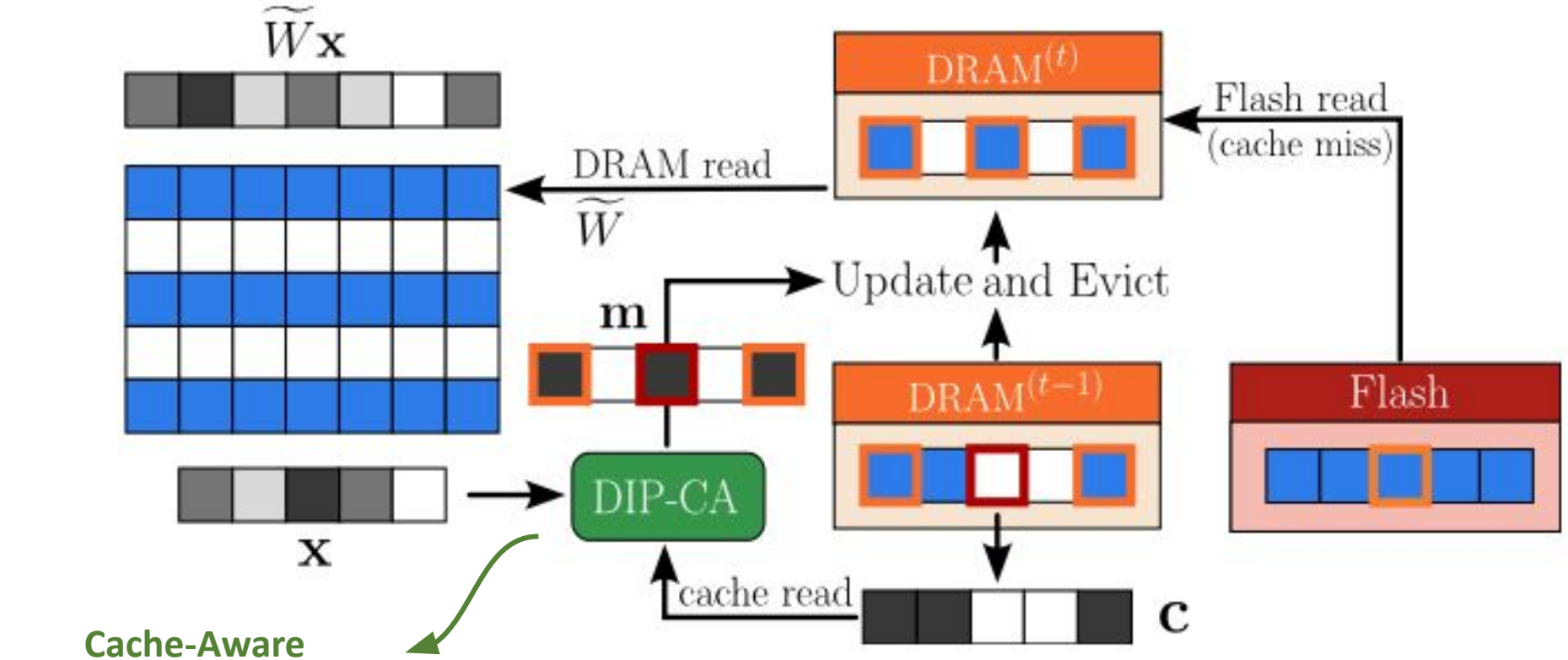
DIP-CA: select weights that have a significant impact on the output and are already in the cache, or weights that are extremely important and worth loading even if not cached.



Cache-Aware Dynamic Input Pruning module

Binary mask indicating which neurons are currently loaded into the DRAM cache.

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Dynamic Width Sparsity

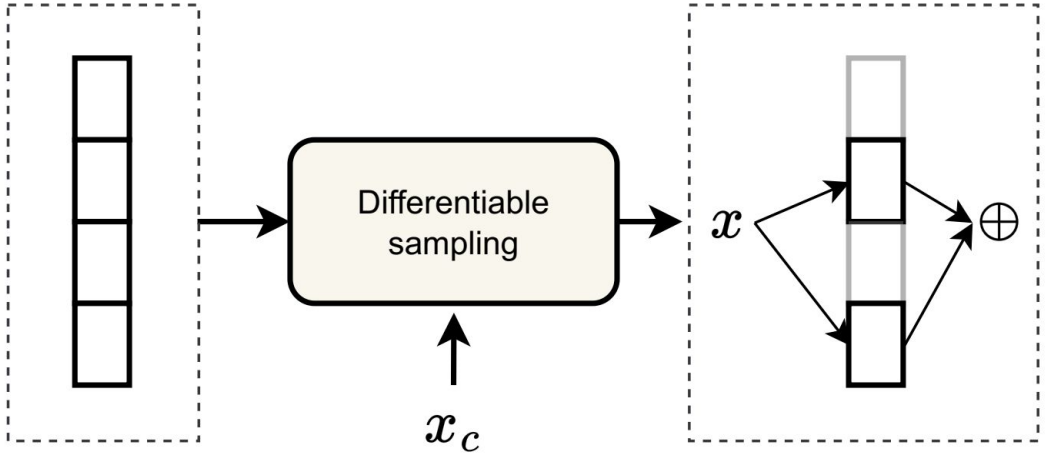

Figure 2: Width sparsity: Different parts of a layer (e.g., *experts*) can be activated based on the conditioning value.

Conditional computation in neural networks: principles and research trends
https://arxiv.org/abs/2403.07965

Dynamic Width Sparsity

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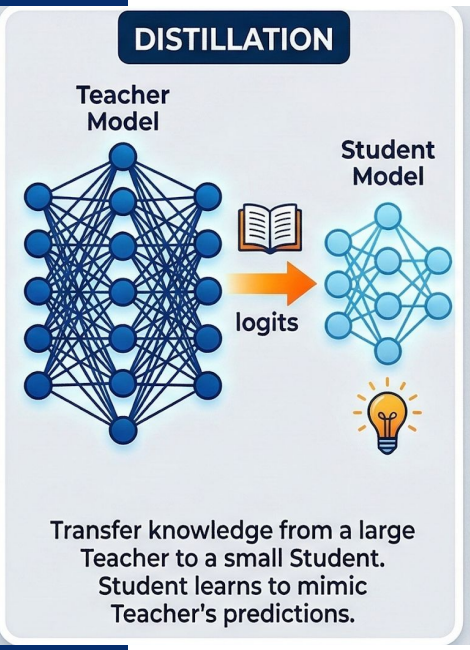
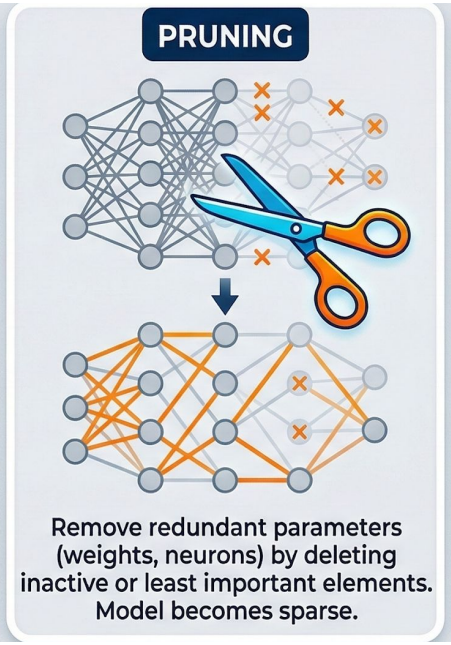
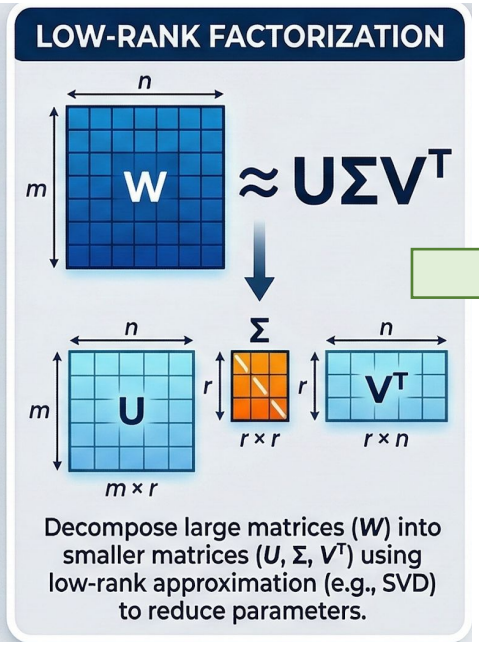
Traditional Approach

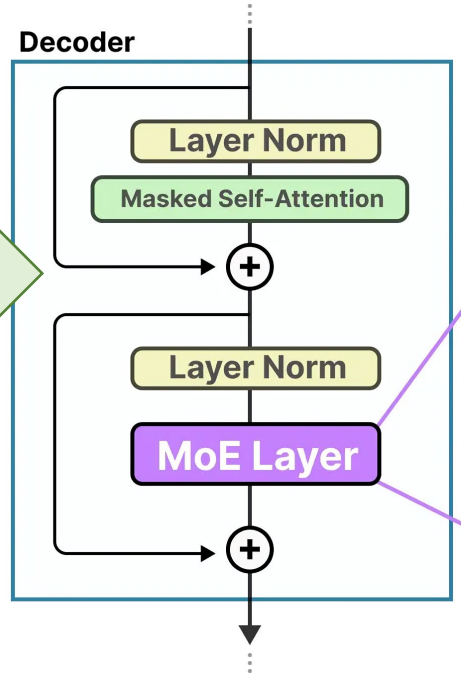
1. **Distillation**: Transferring knowledge from a large "teacher" to a compact "student" model.
2. **Pruning**: Permanently removing unimportant weights, neurons, or connections to reduce size.
3. **Low-rank Factorization**: Using lower-rank matrices to approximate original weights and reduce parameters.

The "Static" Trap: optimized before deployment; once finished, the computation graph remains fixed.

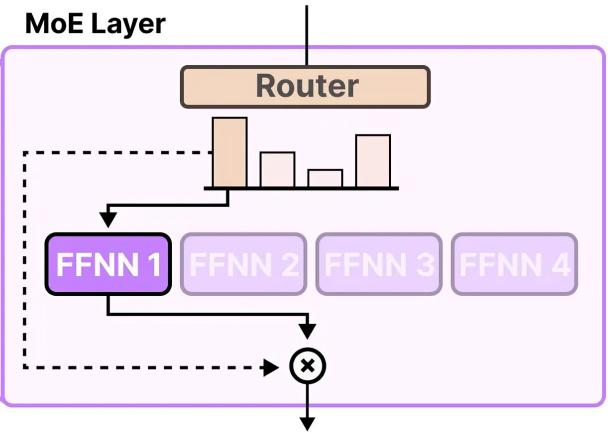








MoE Structure



Transformers: Scaling to Trillion Parameter Models with Simple and Efficient Sparsity

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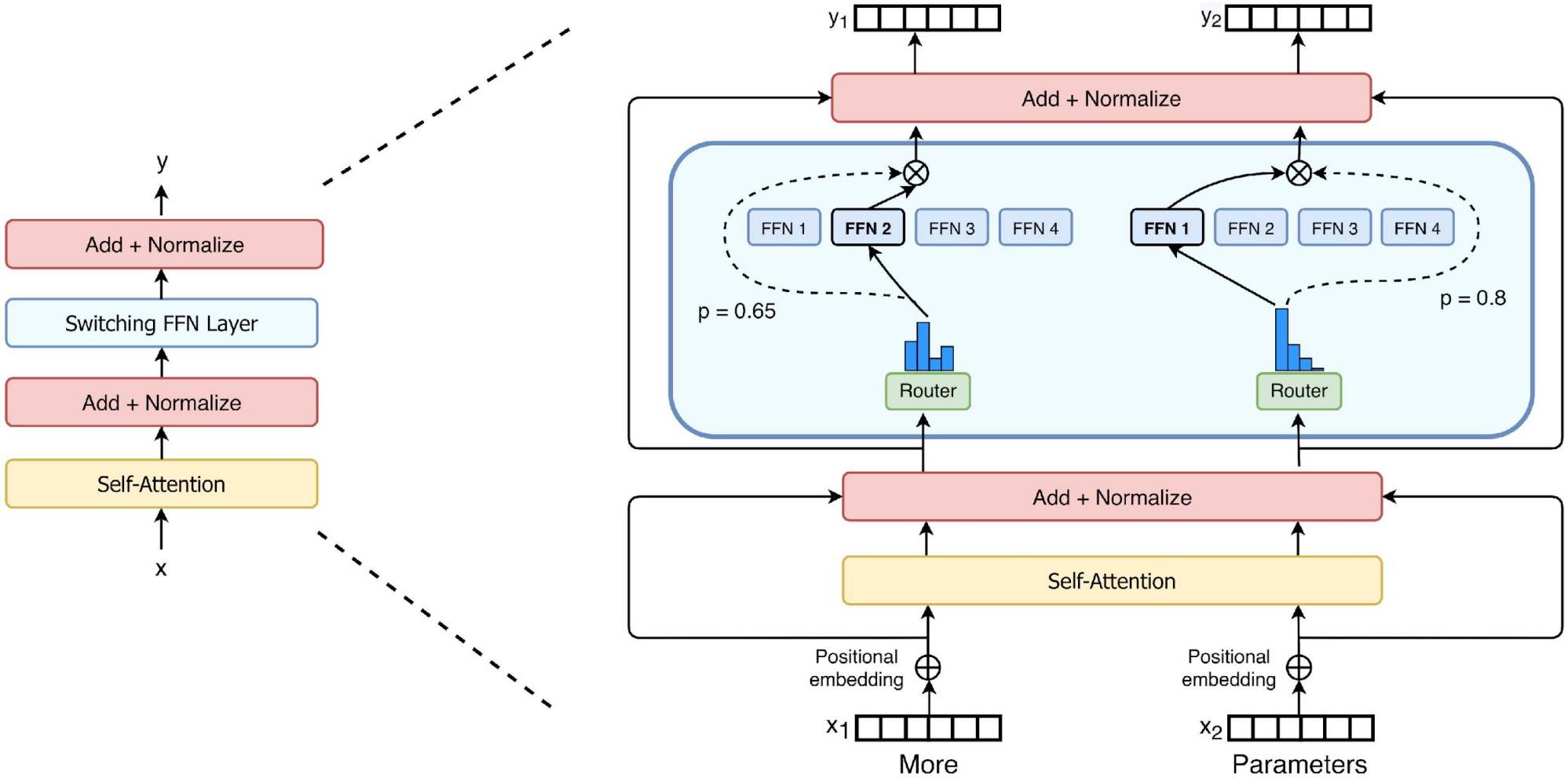
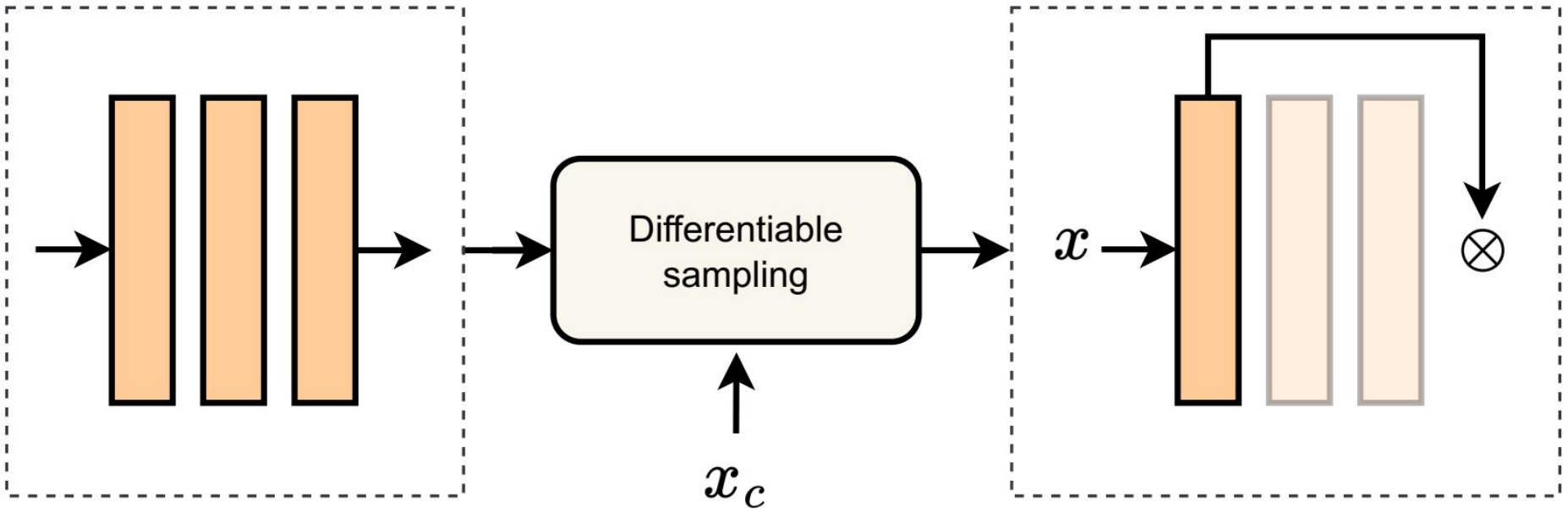


Dynamic Depth Sparsity

1. Pruning and Early Exit
2. Recurrence and Looped Architectures
3. Dynamic Routing and Modular Inference



Conditional computation in neural networks: principles and research trends
https://arxiv.org/abs/2403.07965

1. Pruning and Early Exit

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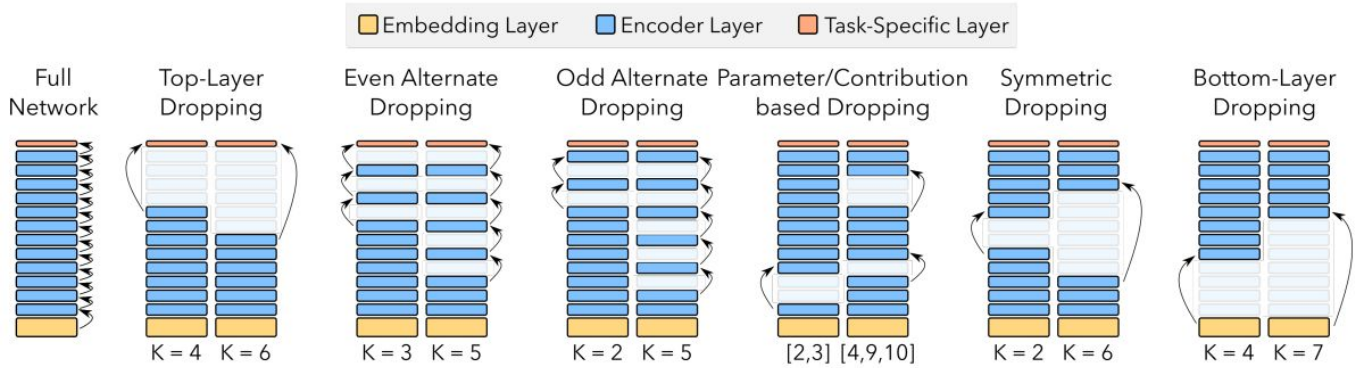
- **Pruning**: remove redundant weights, heads, layers.
- **Early exit**: attach auxiliary classifiers to intermediate layers, where an exit is triggered if the computed entropy falls below a predefined threshold, signifying sufficient confidence in the prediction.



On the Effect of Dropping Layers of Pre-trained Transformer Models

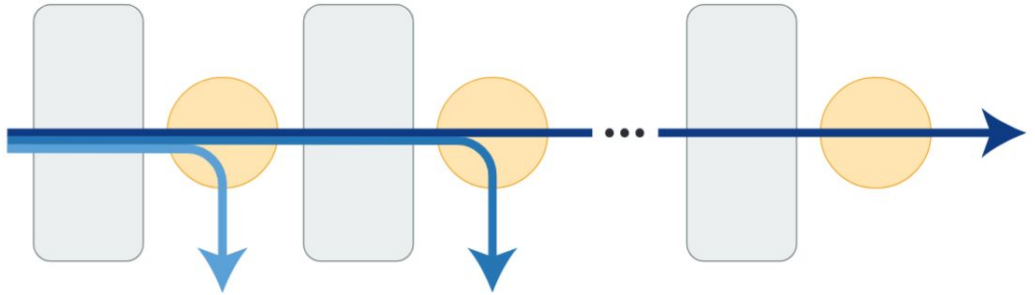


Figure 1: DeeBERT model overview. Grey blocks are transformer layers, orange circles are classification layers (off-ramps), and blue arrows represent inference samples exiting at different layers.

DeeBERT: Dynamic Early Exiting for Accelerating BERT Inference

LayerSkip: Enabling Early Exit Inference and Self-Speculative Decoding
(ACL 2024)

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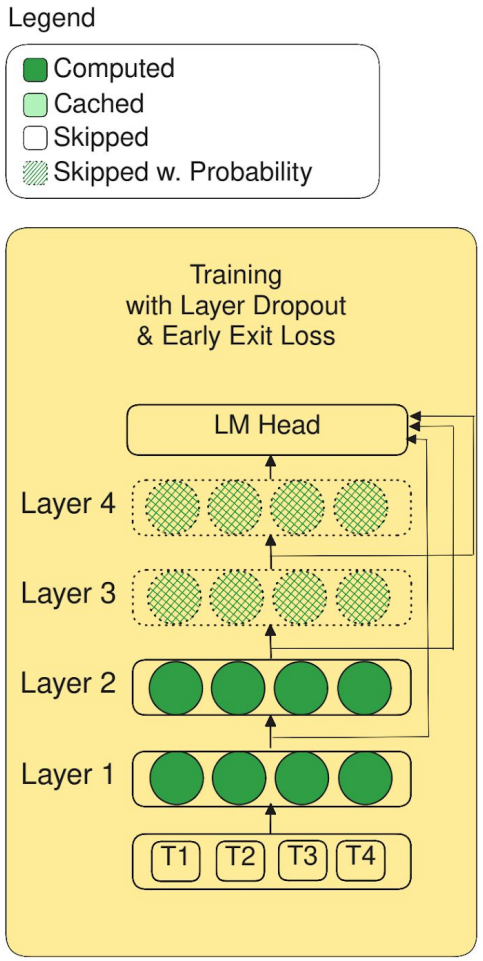
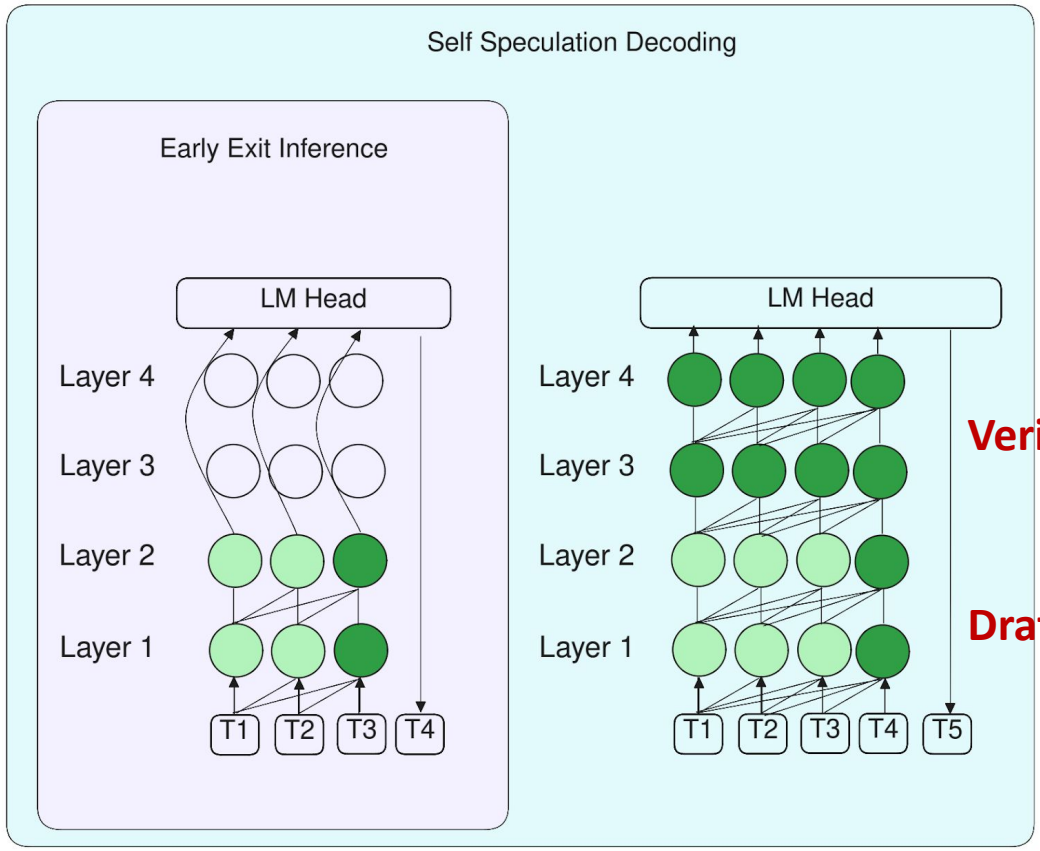


Train using Layer Dropout + Early Exit Loss....



... enables inference with subset of layers with higher accuracy...

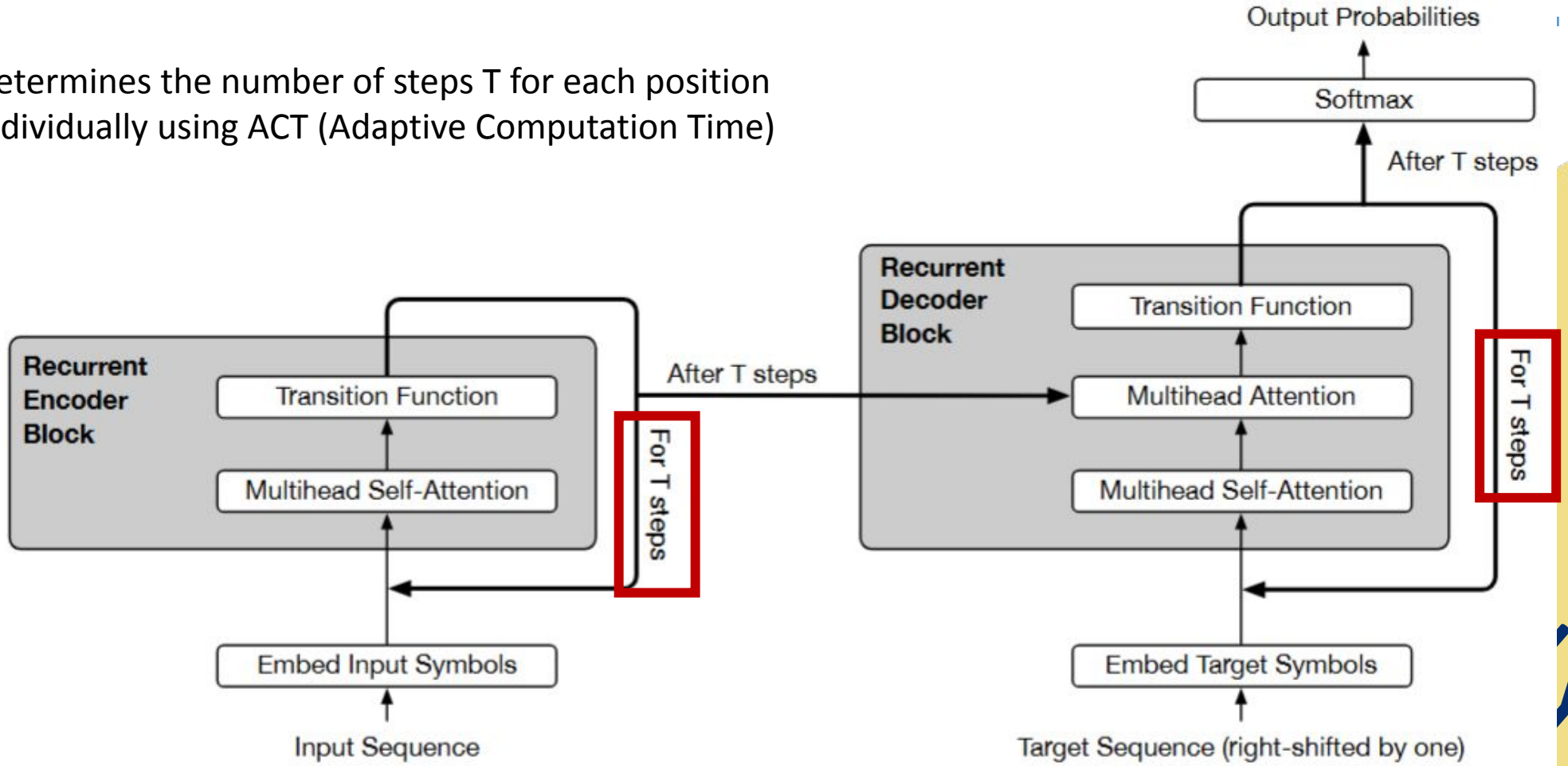
... and we can improve accuracy by verifying and correcting with remaining layers

Verification Stage

Draft Stage

2. Recurrence and Looped Architectures

- Determines the number of steps T for each position individually using ACT (Adaptive Computation Time)



Universal Transformers
https://arxiv.org/abs/1807.03819

3. Dynamic Routing and Modular Inference

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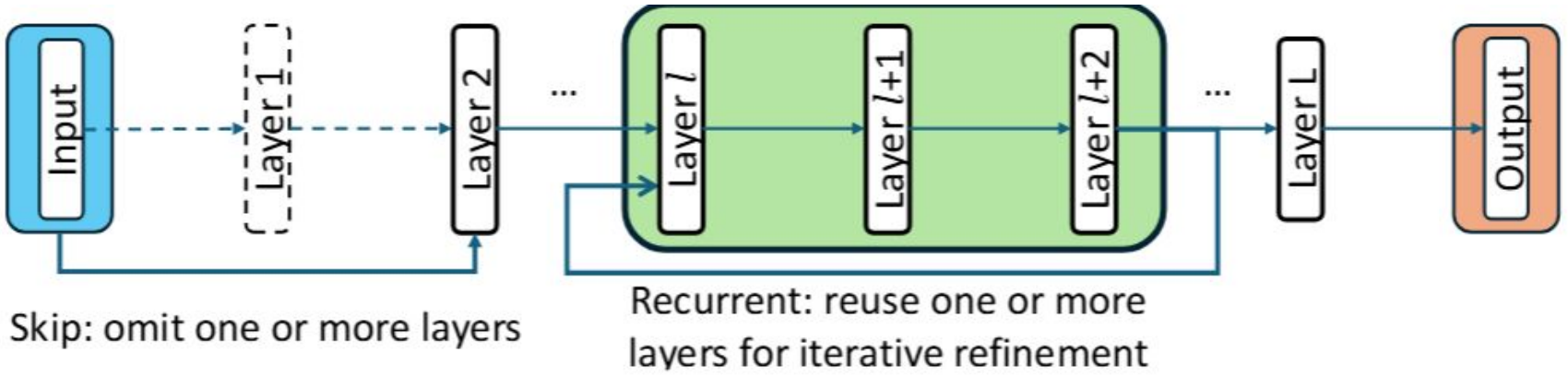

Figure 1: **Test-time layer composition search space** for CoLa. Starting from the original forward path, each input can dynamically skip or recurrently reuse any layer(s) to construct a custom Chain-of-Layers (CoLa). This joint space enables both layer pruning and recurrence, supporting fast-slow depth adaptation and dynamic architecture generalization from a pretrained model without any finetuning.

Skip a Layer or Loop it? Test-Time Depth Adaptation of Pretrained LLMs
https://arxiv.org/abs/2507.07996

Dr.LLM: Dynamic Layer Routing in LLMs
(ICLR 2026)

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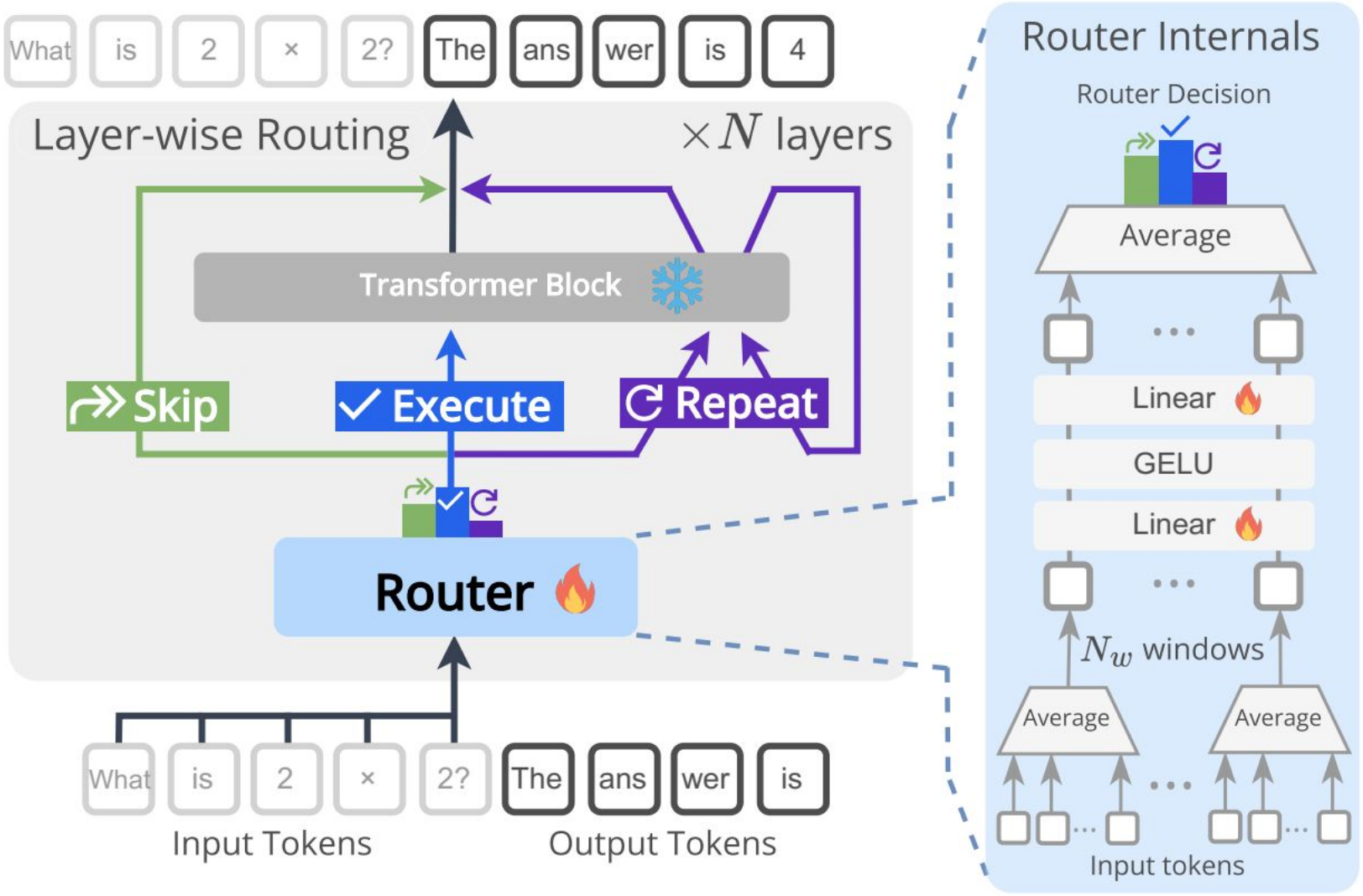

Windowed Mean Pooling for hidden state of input tokens

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Dr.LLM: Dynamic Layer Routing in LLMs
(ICLR 2026)

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- Improving accuracy while simultaneously saving layers (i.e., reducing computation)

Method	Accuracy ↑	Retrofit	Cheap I	Cheap T	LLM ❄
CoLa	✓	✓	✗	✗	✓
Mixture of Depths	✗	✗	✓	✗	✗
Universal Transformer	✓	✗	✗	✗	✗
LLM-Pruner	✗	✓	✓	✗	✗
Mixture of Experts	✓	✗	✓	✗	✗
Mixture of Recursions	✓	✗	✗	✗	✗
LayerSkip	✗	✓	✓	✗	✓
ShortGPT	✗	✗	✓	✗	✗
MindSkip	✗	✓	✓	✗	✓
FlexiDepth	✗	✓	✗	✗	✓
Dr.LLM (Ours)	✓	✓	✓	✓	✓

Recurrence and Looped Architectures
Pruning & Early Exit
Dynamic Width Sparsity

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Dr.LLM: Dynamic Layer Routing in LLMs
(ICLR 2026)

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Limitations (from Hardware View)

- **Dynamic control flow**: Path varies for every prompt, making static compiler optimization and kernel fusion difficult.
- **Irregular Memory Access**:
  - Breaks cache prefetching and data locality when different tasks in a batch skip different layers.
  - Leads to suboptimal throughput on vectorized hardware (GPUs/TPUs).
- **Execution Overhead**: The cumulative latency of Router MLPs and Windowed Pooling can offset the time saved from skipping layers if not carefully implemented.
- **Per-Prompt Constraints**: Requires sequence-level consistency to maintain KV-cache efficiency

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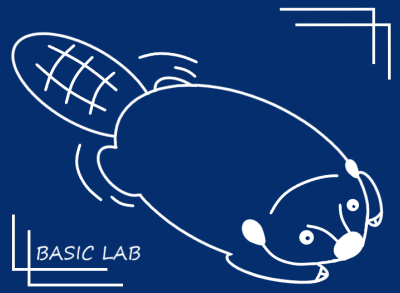
Proposed Research



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Proposed Research 1: Hardware-Friendly Highway Strategy

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- **Problem of Dynamic Routing:**
  - Input-dependent paths cause Warp Divergence and irregular memory access on GPUs.
  - Frequent jumps lead to Instruction Pipeline flushes and Cache Misses.
- **Goal:** Efficient dynamic routing (Dr. LLM) with hardware-friendly highway.
- **Defining "Hot Highways":**
  - Identifying frequent contiguous layer combinations (e.g., Block 1-2-3).
  - Converting unpredictable jumps into **Input-Agnostic Execution Paths** at compile-time

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Proposed Research: Hardware-Friendly Highway Strategy

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Step 1. Stability-Stability-Aware Structure Mining

- Four Metrics for Highway Discovery:
  - **Support**: Frequency of a specific routing pattern (e.g., Skip Layer 7-9) in the training set.
  - **Skip Variance**: Quantifying cross-prompt instability; low variance indicates a candidate for compile-time fixation.
  - **Accuracy Drop**: Measuring the performance impact of forcing a dynamic layer into a static state.
  - **Hardware Contiguity**: Identifying continuous blocks (e.g., 1-2-3) suitable for Operator Fusion.

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Proposed Research:
Hardware-Friendly Highway Strategy

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Step 2. Compiler & Hardware Mapping Pipeline

- **Path Profilling:** Extracting path locality for specific tasks.
- **Graph Transformation:** Converting conditional branches into specialized **"Fast-path" Sub-graphs**.
- **Kernel Fusion:** Merging consecutive Highway blocks into a single CUDA/Triton kernel to minimize VRAM I/O.
- **Static Path Optimization:** Establishing dedicated physical **Data Paths** (in FPGA/ASIC) for high-frequency routes to minimize unit-to-unit latency.
- **Block Clustering:** Mapping frequently co-executed blocks onto **physically adjacent Processing Elements (PEs)** to maximize local SRAM reuse.
- **Static SRAM Allocation:** Pre-loading Highway weights/activations into SRAM to eliminate dynamic scheduling bubbles.

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Extension to MoE Structure

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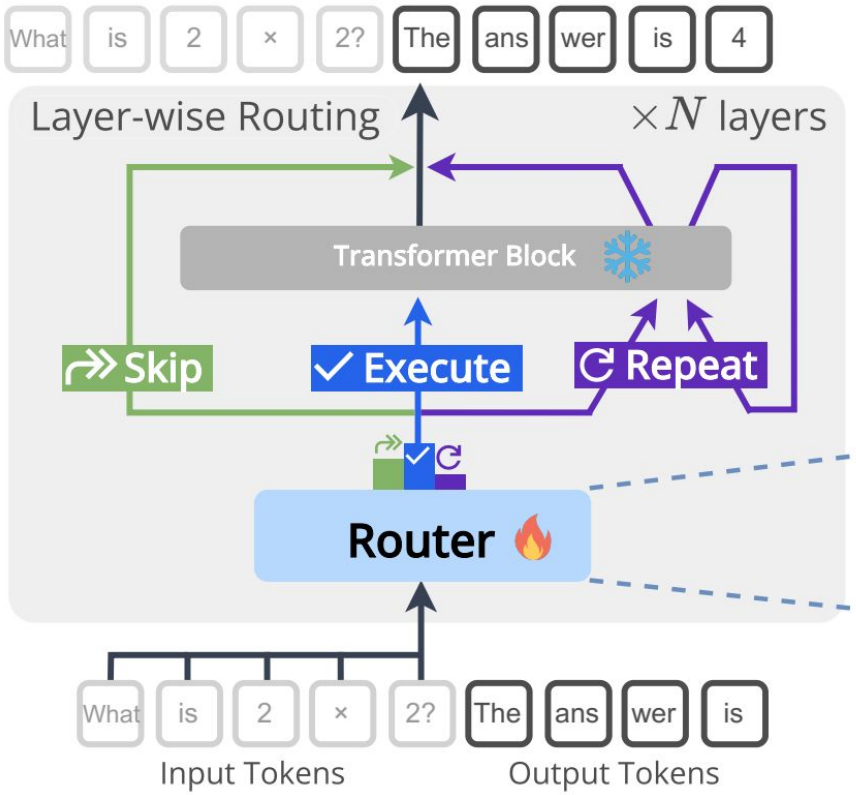
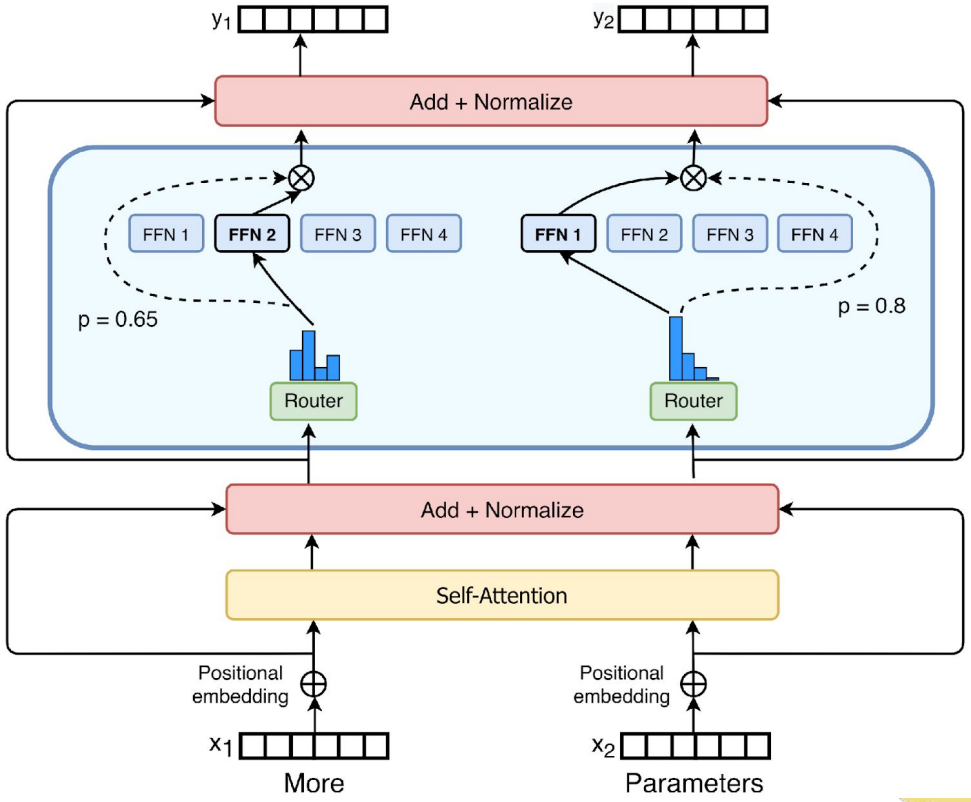
Layer-wise Routing



Expert-wise Routing



Proposed Research 2:
MoE Speculative Decoding

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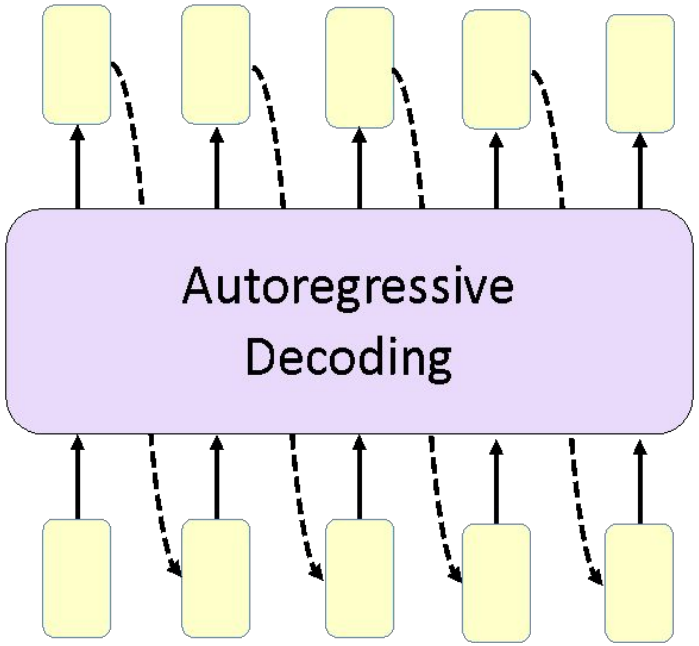

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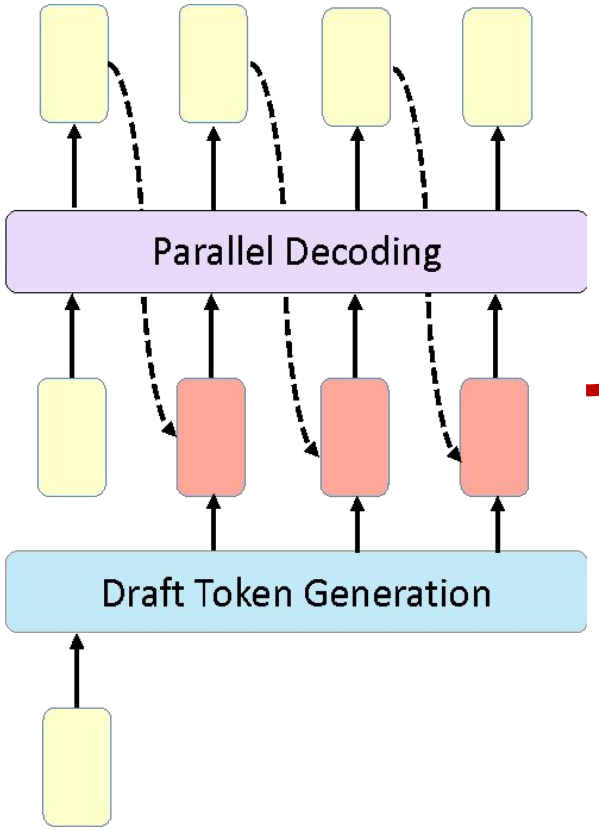

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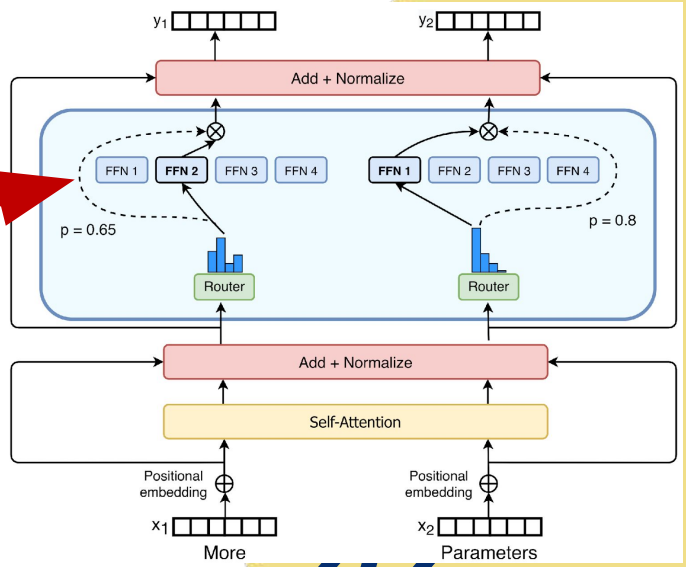
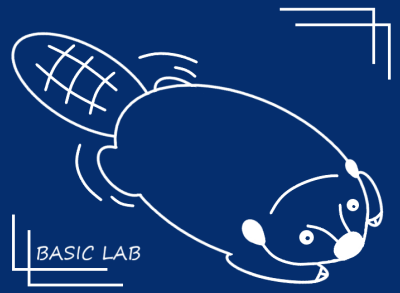


THANK YOU FOR LISTENING

Big data Analytics and Social Intelligent Computing



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